10.4 A Molecule and a Piston

WORKSHEET

The question

STEP I 2002, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s1-q10

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself A ball moving at speed w hits a wall that is moving towards it at speed u, and bounces elastically. What is its speed afterwards? (Think in the wall's frame.)

STAGE 2 A recurrence for the length

Ask yourself Between the nth and $(n+1)$th collisions, the particle goes from the piston to the end and back. How far does it travel? How far does the piston move?

STAGE 3 Telescoping

Ask yourself Write d_n as d_1 times a product of $n-1$ such fractions. What cancels?

Try this next

1. **The invariant exactly.** Show that $v_n d_n - (v+u)d_1 = \frac{u(v+u)d_1}{v+(2n-1)u}$, and deduce that $v_n d_n \to (v+u)d_1$ as $n \to \infty$.

2. **Pulling the piston out.** Repeat the question with the piston withdrawn at speed $u < v$. Show that the particle slows down, and find $v_n d_n$. (Expanding gases cool.)

3. **Energy.** In the case $v = u$, the molecule's speed grows without limit as the piston approaches the end. Find the average force on the piston during the nth cycle (change of momentum divided by t_n) and show that it grows like n^3. What does this say about the work needed to push the piston all the way in?

4. **A bouncing ball.** A ball bounces elastically on a floor under gravity, and the floor is raised slowly. Guess an adiabatic invariant for this motion. (*Hint:* the analogue of vd is the average speed times the height of the bounce, or equivalently the energy times the time between bounces.)

Further reading

- **Richard Feynman, Robert Leighton & Matthew Sands, *The Feynman Lectures on Physics*, Vol. I** (free online), Chapter 39, "The kinetic theory of gases". The same piston argument, extended to three dimensions.

- **Stephen Blundell & Katherine Blundell, *Concepts in Thermal Physics*** (Oxford University Press). Adiabatic processes and kinetic theory at first-year level.
- **Lev Landau & Evgeny Lifshitz, *Mechanics*** (Butterworth-Heinemann), §49. A concise treatment of adiabatic invariants, for later.

The full guided solution, and what the question leads to at university, is in Section 10.4:
https://beyondstep.keithmcnulty.org/ch10/piston.html